Grand Challenge

Cloud-Free Line-of-Sight Prediction for
Optical Ground Station Networks

Part A: Cloud Segmentation & Network Availability
Part B: Business Model

QOSMIC SPACE — Founder's Office Interview
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Part A1: Cloud Segmentation — Model & Multi-Sensor Extension

Architecture

The prototype uses a U-Net with EfficientNet-B0 encoder, trained on the 38-Cloud dataset (Landsat 8, RGB bands, geographic scene-ID split: 12 train / 3 val / 3 test). Training uses AMP, Dice+Focal loss combination, cosine annealing schedule, over 50 epochs with checkpointing.

Real Results (Not Cherry-Picked)

Metric	Value
Validation mIoU (geographic split)	82.2%
Test mIoU (geographic split)	44.0%
Cloud precision	97.6%
Cloud recall	34.6%
Expected Calibration Error (ECE)	0.27

The 44% test score reflects honest geographic generalization to unseen regions — the model was trained on specific scenes and tested on entirely different geographies. This is a harder but more realistic evaluation than random patch splitting.

Multi-Sensor Extension Architecture

To extend the single-sensor prototype to multiple satellite sources (Sentinel-2, INSAT-3D, Landsat 8/9), we insert a spectrum-aware projection layer before the encoder:

```
Input: [B, C, H, W] where C varies per sensor
       (4 for Landsat 8, 13 for Sentinel-2, 6 for INSAT-3D)
```

```
    │
    ▼
Spectral Projection:
  Per-band 1×1 Conv → LayerNorm → ReLU
  Each band mapped to D-dimensional feature
  Output: [B, C, D, H, W]
    │
    ▼
Band Pooling:
  Mean over C dimension → [B, D, H, W]
  Invariant to number of input bands
    │
    ▼
1×1 Conv → [B, 3, H, W] (match EfficientNet-B0 input)
    │
    ▼
Encoder (EfficientNet-B0, ImageNet pretrained)
```

Each band carries a spectral position embedding — a learned vector tied to the central wavelength. Bands at similar wavelengths (e.g., Sentinel-2 band 4 at 665nm and INSAT-3D visible at 650nm) get similar embeddings. The encoder learns that "bands at this spectral position behave like this for cloud detection."

Handling Missing Bands

A sensor without thermal bands (Sentinel-2) still produces reasonable predictions because the band pooling averages over whatever bands are available. Training with random band dropout (10% probability per band) forces the model to not become dependent on any single band. When SWIR bands are absent, the model learns to output lower confidence — appropriate since thin cirrus is invisible without SWIR.

Expected Failure Modes

- **Resolution mismatch:** 100× gap between Sentinel-2 (10m) and INSAT-3D (1km). Mitigation: train on S2 downsampled to 1km for joint training.
- **Band absence:** No SWIR means thin cirrus detection suffers. The model calibrates confidence downward when SWIR bands are absent.

- **Temporal misalignment:** S2 passes every 5 days; INSAT-3D every 15 minutes. Temporal consistency across INSAT frames serves as a self-supervised signal.

Part A2: Network Availability & Scheduling

Recommended Station Locations

Station	Location	Annual Clear	Rationale
Leh, Ladakh	34°N, 77°E	~74%	Cold desert, 300+ clear days/year, high altitude (3500m), low turbulence
Jodhpur, Rajasthan	26°N, 73°E	~78%	Thar Desert fringe, monsoon is short and weak
Challakere, Karnataka	14°N, 76°E	~46%	QOSMIC's existing site, semi-arid interior
Sriharikota, AP	14°N, 80°E	~50%	ISRO launch site, coastal but operable
Shillong, Meghalaya	26°N, 92°E	~40%	NE India, different weather system from all others

Network Availability Results

Configuration	Annual Availability
Single station (Challakere only)	46%
5-station network	>99%
30-day simulation: passes with no clear station	0
30-day simulation: conflicts resolved by routing	46 (15%)

Station Load Distribution (30-day simulation)

Station	Share of Passes
Jodhpur	39%

Leh	35%
Shillong	18%
Sriharikota	7%
Challakere	1%

Key Assumptions

- Monthly climate averages from MERRA-2 reanalysis (NASA) are representative
- 1 monitoring engineer per 3-5 stations
- Leh winter hardware (cold-rated optics) can be sourced within budget
- Analysis uses daily-average cloud data; hourly ERA5 would enable finer scheduling

Part B: Business Model

Spreadsheet Structure (6 Tabs)

The business model is a living spreadsheet with 40+ labeled input cells (blue). Change any assumption and the entire model updates. Available at: *Google Sheets link in GC README*

Tab	Contents
Assumptions	CapEx/OpEx per station, pricing, station count, cloud availability, customer growth
Cloud Model	5-station availability, scheduler simulation results, Part A data feeds directly
Unit Economics	Revenue per pass, cost per GB, breakeven at various utilization levels
5-Year P&L	5 revenue streams, 6 cost lines, EBITDA, monthly granularity for Year 1
Breakeven Sensitivity	Customers × ARPU matrix, price sensitivity at 20/40/60/80% utilization
Investor Summary	Auto-updating dashboard: key metrics, raise amount, dilution, comparable transactions

Part A → B Integration

The Cloud Model tab uses the same 5 stations and >99.5% network availability from Part A's analysis. Station deployment schedule matches Part A recommendations. CapEx per station ($200K) comes from Part A's BoM research. This is the key cross-link the PDF evaluates: cloud-diversity analysis from Part A directly feeds the network economics in Part B.

Revenue Streams

Stream	Timeline	Pricing	Year 5 Revenue

OGS-as-a-Service	Year 1+	$1/GB	$4.2M
Partner Network Commissions	Year 1+	$0.20/GB	$657K
NQM Government Contracts	Year 1+	Phased deliverables	$150K
ARGUS Hardware Sales	Year 4+	$350K/unit	$700K
ZAPHOD Terminal Sales	Year 2+	Per unit	$0

Financial Summary

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue	$337K	$1.0M	$2.1M	$3.8M	$5.7M
EBITDA	-$221K	-$82K	$453K	$1.2M	$1.99M
Stations owned	1	2	3	4	5

Series A Raise

Parameter	Value
Target raise	$4-6M
Pre-money valuation	$15-25M
Dilution (midpoint)	~18%
Breakeven	Year 3

Use of Funds ($5M Midpoint)

Category	Amount	Detail
Station CapEx (stations 2-5)	$1.8M	4 stations × ~$200K + contingency
Team expansion (8→15)	$1.2M	18 months payroll at $30K blended avg
Partner network growth	$0.4M	BD + integration for 20+ stations
Operating expenses	$0.4M	Power, fiber, maintenance, leases
Office + overhead	$0.4M	Bangalore scaling, legal, compliance

Sales & BD	$0.3M	Customer acquisition, conferences
Working capital buffer	$0.5M	6-month runway for NQM payment delays
Total	**$5.0M**	

Top 3 Risks & Mitigations

1. Customer concentration. Year 1 revenue is entirely NQM contracts ($200K). Mitigation: NQM has government commitment letters. $0.5M working capital buffer. Partner network provides immediate bookable product.

2. Station deployment delays at remote sites. Leh (-30°C winter) and Shillong (2000mm monsoon) pose construction challenges. Mitigation: Challakere deployment documented the build procedure. Leh scheduled for dry window. Backup sites at Nagpur and Hyderabad identified.

3. Optical link reliability at scale. Solo-station availability ranges from 46% (Challakere) to 78% (Jodhpur). Dropping one station still gives 97%+ availability. Published probabilistic link budget model (arXiv:2507.20908) provides framework for dynamic pass scheduling.